
Closed-Loop Imitation Learning for Battery Control in Distribution Systems

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Abstract

Battery Energy Storage Systems (BESS) in distribution networks are typically dispatched by a fixed-window, rolling-horizon linear program (LP) Energy Management System (EMS) re-solved every hour. While near-optimal, this LP is a computational bottleneck when embedded inside outer planning loops that require hundreds of co-simulation evaluations. We apply imitation learning—specifically Behavioral Cloning (BC) and Dataset Aggregation (DAGger)—to train a Random Forest policy that mimics the EMS expert from inputs available at each hourly decision boundary: battery state of charge (SOC, kWh) and the rolling 24-hour price forecast ($\mathbf{c} \in \mathbb{R}^{24}$, \$/kWh). The learned policy replaces only the hourly EMS dispatch; a local voltage-deadband active-power droop layer runs every 5 minutes and enforces hard SOC clamping identically in all setups. We evaluate on the IEEE 34-bus unbalanced distribution feeder in a high-fidelity OpenDSS/HELICS co-simulation over 40-day runs at three price-noise levels ($\sigma \in \{0.004, 0.04, 0.4\}$ \$/kWh). BC matches LP revenue exactly at low noise and captures 89.8% at high noise while achieving *lower* voltage RMSE than LP. DAGger further reduces setpoint MSE by 17% in the hardest scenario. Voltage quality is essentially unchanged across all controllers, indicating that the droop layer provides a robust safety floor that partially mitigates distribution shift.

1 Introduction

The rapid growth of distributed energy resources—particularly rooftop photovoltaics (PV) and battery energy storage systems (BESS)—is transforming electric power distribution networks [2]. BESS assets can absorb excess solar generation during midday overvoltage events and discharge during evening peak-load periods, providing energy arbitrage revenue. The time-coupled nature of state-of-charge (SOC) dynamics makes real-time BESS dispatch a challenging sequential decision-making problem.

The dominant approach is optimization-based: at the start of each hour, a rolling-horizon EMS solves an LP over a fixed 24-hour look-ahead window, computing an active-power setpoint P_h^{pref} that maximizes energy arbitrage subject to SOC feasibility. Between hourly re-solves, a local droop layer handles fast voltage regulation. Although each LP solve completes in milliseconds, the EMS is frequently embedded in outer planning loops searching over hundreds of BESS siting and sizing candidates, each requiring 24 hourly re-solves across multiple seasonal days. Replacing the LP with a single forward pass through a trained policy eliminates this bottleneck entirely.

This work makes three contributions: (i) an empirical study of imitation learning for BESS dispatch in unbalanced distribution networks, providing a novel physical domain for testing distribution shift theory; (ii) a reproducible OpenDSS/HELICS co-simulation testbed on the IEEE 34-bus feeder with price-noise stress scenarios; (iii) a controlled comparison of BC and DAGger against the LP-EMS

expert. Among four architectures evaluated (MLP, LSTM, LSTM with auxiliary task, Random Forest), Random Forest performed best, underscoring the value of tabular structure when SOC and a 24-slot price vector are the primary decision inputs.

2 Related Work

Behavioral cloning. The foundational example of BC is ALVINN [6], which trained a three-layer network to map camera images to steering commands, demonstrating end-to-end supervised learning on expert demonstrations—and exposing the central weakness: performance degrades away from expert-visited states. Our work applies the same paradigm to BESS dispatch, with the EMS as expert and (SOC, price forecast) as the observation.

Distribution shift and DAgger. Ross, Gordon, and Bagnell [7] formalized BC’s compounding error problem and proposed DAgger, proving that iterative dataset aggregation reduces performance degradation from $O(T^2\epsilon)$ to $O(T\epsilon)$. We test whether this advantage transfers to a physical domain where hard SOC bounds and droop enforcement constrain state divergence.

Optimization-based BESS dispatch. Fortenbacher, Mathieu, and Andersson [2] present a two-stage MPC scheme for distributed BESS using a linearized optimal power flow formulation that jointly manages voltage and battery degradation. Their LP-based MPC is the direct predecessor of the EMS oracle we use; unlike their approach, our learned policy requires no network model or load forecast at inference time.

Learning-based voltage control. Liu and Wu [4] propose a two-stage deep RL framework for inverter-based Volt-VAR control, using adversarial offline pre-training followed by online fine-tuning. RL requires careful reward shaping and online safety monitoring; our approach uses purely offline BC training with DAgger querying the expert at learner-visited states.

Learning to solve optimal power flow. Zamzam and Baker [8] train a network on AC OPF solutions to produce near-optimal dispatch in milliseconds. The key distinction is that their problem is a single-shot regression; ours is sequential, so the policy’s actions shift future SOC, making distribution shift a central concern.

ML-based battery energy management. Nakabi and Toivanen [5] apply DQN to microgrid battery management. Their agent requires tens of thousands of environment interactions and does not incorporate voltage regulation; our approach generates high-quality training data efficiently by imitating the EMS oracle.

3 Problem Statement

Notation. We use c_t (\$/kWh) for electricity prices and f_θ for the learned dispatch policy to avoid conflict.

At each hourly decision boundary h , the observable available to the controller is:

$$o_h = [\text{SOC}_h, c_0, c_1, \dots, c_{23}] \in \mathbb{R}^{25}, \quad (1)$$

where $\text{SOC}_h \in [\text{SOC}^{\min}, \text{SOC}^{\max}]$ is the battery energy level (kWh) read from the droop federate at the hourly boundary, and $(c_0, \dots, c_{23})$ is the rolling 24-step price forecast in \$/kWh—the same vector the EMS LP uses, with optional persistent Gaussian perturbations on far-horizon slots (see Section 4.2). The controller outputs a scalar active-power setpoint $P_h^{\text{ref}} \in [-\bar{P}, \bar{P}]$ (kW; positive = discharge, inject to grid) that is *held constant* by the droop federate for all 12 five-minute sub-steps of hour h .

The imitation learning goal is to find $f_\theta : o_h \mapsto P_h^{\text{ref}}$ minimizing:

$$\min_{\theta} \mathbb{E}_{o_h \sim d_{f_\theta}} \left[(f_\theta(o_h) - f^*(o_h))^2 \right], \quad (2)$$

where f^* is the EMS oracle and d_{f_θ} is the state distribution induced by the learner. BC minimizes this over d_{f^*} (expert’s distribution); DAgger iteratively expands the training distribution toward d_{f_θ} , reducing the compounding error bound from $O(T^2\epsilon)$ to $O(T\epsilon)$ [7].

4 Method

4.1 Expert Oracle: Fixed 24-Step Rolling-Horizon LP

At the start of each hour h , the EMS solves an LP over a **fixed 24-step horizon** (non-shrinking; re-rolls each hour, MPC-style):

$$\max_{\{P_t\}_{t=0}^{23}} \sum_{t=0}^{23} c_t P_t, \quad \text{where } c_t = c_{(h+t) \bmod 24}, \quad (3)$$

subject to SOC dynamics (1-hour steps, efficiency η):

$$\text{SOC}_{t+1} = \text{SOC}_t - \eta P_t, \quad P_t \in [-\bar{P}, \bar{P}], \quad \text{SOC}_t \in [\text{SOC}^{\min}, \text{SOC}^{\max}], \quad (4)$$

where $c_{(h+t) \bmod 24}$ is the forecast price at LP horizon step t (wrapping the 24-hour CSV profile), and units are such that $c_t \cdot P_t$ gives \$/h. Only the first-slot solution $P_h^{\text{ref}} = P_0$ is dispatched; the LP is re-solved at $h+1$ from the measured SOC.

BESS parameters (from `common.py`): $\bar{P} = 500$ kW, $\text{SOC}^{\min} = 200$ kWh, $\text{SOC}^{\max} = 1800$ kWh, $\text{SOC}^{\text{init}} = 1000$ kWh. The droop federate uses $\eta = 0.95$ (ETA_CH); the LP's internal efficiency is defined in the `rhemes` solver module.

4.2 Price Noise Model

The EMS (and BC policy) receive a rolling 24-step price vector. For each absolute simulation hour $a = h + t$:

$$c_t^{\text{LP}} = \begin{cases} c_{a \bmod 24}^{\text{CSV}} & \text{if } a < 24 \text{ (first day, no noise),} \\ c_{a \bmod 24}^{\text{CSV}} + \varepsilon_a & \text{if } a \geq 24, \end{cases} \quad (5)$$

where $\varepsilon_a \sim \mathcal{N}(0, \sigma^2)$ is drawn *once* when absolute slot a first appears in a horizon and then *fixed* for the remainder of the simulation (no daily reset, no re-draw). This models forecast uncertainty that accumulates over multi-day runs without compounding from prior perturbations.

4.3 Droop Layer (Voltage-Deadband Active-Power Droop)

The droop federate runs every 5 minutes. It reads P_h^{ref} (constant within the hour) and the post-power-flow voltage magnitude V_{pu} at bus 830 phase C. A deadband active-power correction is computed:

$$\delta P = \begin{cases} -K (V_{\text{pu}} - v_{\text{hi}}) & V_{\text{pu}} > v_{\text{hi}}, \\ K (v_{\text{lo}} - V_{\text{pu}}) & V_{\text{pu}} < v_{\text{lo}}, \\ 0 & \text{otherwise,} \end{cases} \quad (6)$$

with reference $V_{\text{ref}} = 1.00$ pu, half-deadband $\Delta_V = 0.01$ pu ($v_{\text{hi/lo}} = V_{\text{ref}} \pm \Delta_V$), and gain $K = 2000$ kW/pu. The dispatched power and SOC update are:

$$P_{\text{act}} = \text{clip}(P_h^{\text{ref}} + \delta P, -\bar{P}, \bar{P}), \quad (7)$$

$$\text{SOC} \leftarrow \text{clip}(\text{SOC} - \eta P_{\text{act}} \frac{5}{60}, \text{SOC}^{\min}, \text{SOC}^{\max}), \quad (8)$$

where $\eta = 0.95$ and $5/60$ h is the 5-minute step size. This layer is *identical* across LP-EMS, BC, and DAgger setups; only the source of P_h^{ref} changes.

4.4 Co-simulation Environment

We use OpenDSS [1] coupled with HELICS [3] on the IEEE 34-bus unbalanced three-phase radial test feeder. Figure 1 shows the architecture. Four federates are ordered by INT_INDEX_GROUP to eliminate publication lag: *EMS/Policy* (group 0, 60-min), *Droop* (group 1, 5-min), *Network/OpenDSS* (group 2, 5-min), and *Logger* (group 3). Price noise is generated inside the EMS/Policy federate. The BESS is modeled as an OpenDSS Generator (bus 830, 24.9 kV, three-phase, ± 500 kW); all SOC integration occurs in the Droop federate only. The EMS reads only SOC from the Droop federate at each hourly boundary—it has no access to bus voltage.

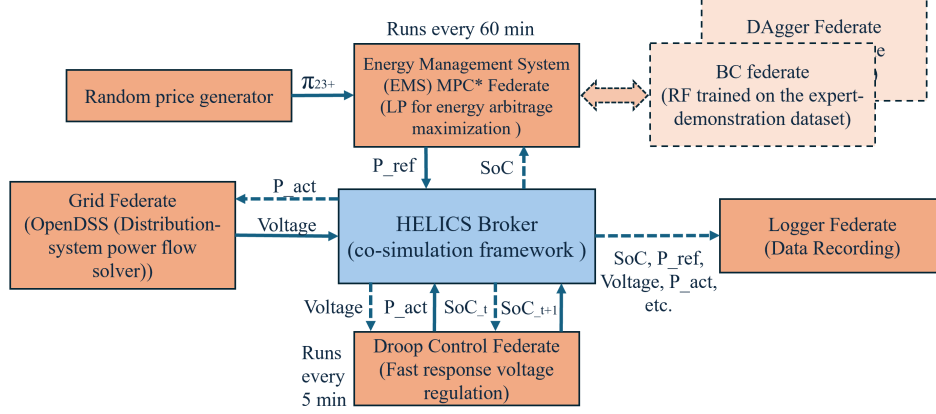


Figure 1: Four-federate HELICS co-simulation on the IEEE 34-bus feeder. The EMS/Policy federate is the only component that differs between LP-EMS, BC, and DAgger setups.

4.5 Behavior Cloning: Observation and Policy

For the tabular Random Forest (RF), the raw 25-dimensional o_h (Eq. 1) is augmented with engineered features: price mean, std, min, max, span; first three and last price slots; first/second half means and their gap; consecutive price differences (mean, std, max, min); and three SOC-price interaction terms ($\text{SOC} \times c_0$, $\text{SOC} \times \bar{c}$, $\text{SOC} \times \text{span}$). The RF outputs P_h^{ref} clipped to $[-\bar{P}, \bar{P}]$, trained by minimizing MSE against EMS labels.

For LSTM models, the 24-price vector is passed as a sequence to a 2-layer LSTM (hidden size 64), while SOC is processed by a separate linear branch (output size 16). Their concatenated output feeds an MLP head producing either one output (immediate P^{ref}) or 24 outputs (LSTM + auxiliary task, predicting the full LP horizon). All models are trained offline on EMS rollouts.

4.6 DAgger

Starting from the BC dataset, DAgger iterates: (1) deploy the current policy f_{θ}^{i-1} in the co-simulation at $\sigma = 0.4$; (2) query the EMS oracle f^* for $P_h^{\text{ref},*}$ at every learner-visited state o_h ; (3) aggregate the new $(o_h, P_h^{\text{ref},*})$ pairs into the training set; (4) retrain f_{θ}^i . High price-noise rollouts are used so the oracle is queried at exactly the out-of-distribution states where BC degrades.

5 Experimental Design

5.1 Domain and Data

All experiments use the IEEE 34-bus three-phase unbalanced radial feeder in OpenDSS/HELICS. A single BESS is placed at bus 830 (phase C, a high-voltage-sensitivity node). Three price-noise levels are tested: $\sigma \in \{0.004, 0.04, 0.4\}$ \$/kWh. Each run spans 40 simulated days at 5-minute resolution (960 h, 11,520 timesteps).

5.2 Hypotheses

H1: BC produces a functional dispatch policy at $\sigma = 0.004$, matching LP revenue and voltage quality. **H2:** Higher σ exposes distribution shift, increasing setpoint MSE between BC and LP. **H3:** DAgger trained at $\sigma = 0.4$ recovers BC’s degradation, achieving lower setpoint MSE and higher revenue than BC alone.

5.3 Evaluation Metrics

Three metrics are reported. **Revenue proxy** is $R = \sum_t c_{h(t)}^{\text{CSV}} P_{\text{act},t} \Delta t_5$, where $\Delta t_5 = 5/60$ h and $h(t)$ is the clock hour of step t ; this evaluates realized economic value at 5-min resolution using the *unperturbed* CSV price. **Voltage RMSE** is $\sqrt{\mathbb{E}[(V_{\text{pu}} - 1)^2]}$ [pu] at bus 830 phase C. **Setpoint MSE** is $\mathbb{E}[(P_{\text{BC}}^{\text{ref}} - P_{\text{LP}}^{\text{ref}})^2]$ [kW²], comparing hourly setpoints of BC and LP under the same conditions.

6 Results and Discussion

6.1 Environment Validation

We first validated the four-federate co-simulation. The LP-EMS converged at every hourly boundary; the droop federate maintained bus-830 voltage near 1 pu at every 5-minute sub-step; SOC remained within [200, 1800] kWh; and the INT_INDEX_GROUP ordering ensured P^{ref} was published before the Droop federate consumed it, eliminating one-step lag artifacts. This was a non-trivial engineering milestone given the multi-federate, time-coupled, physics-constrained architecture.

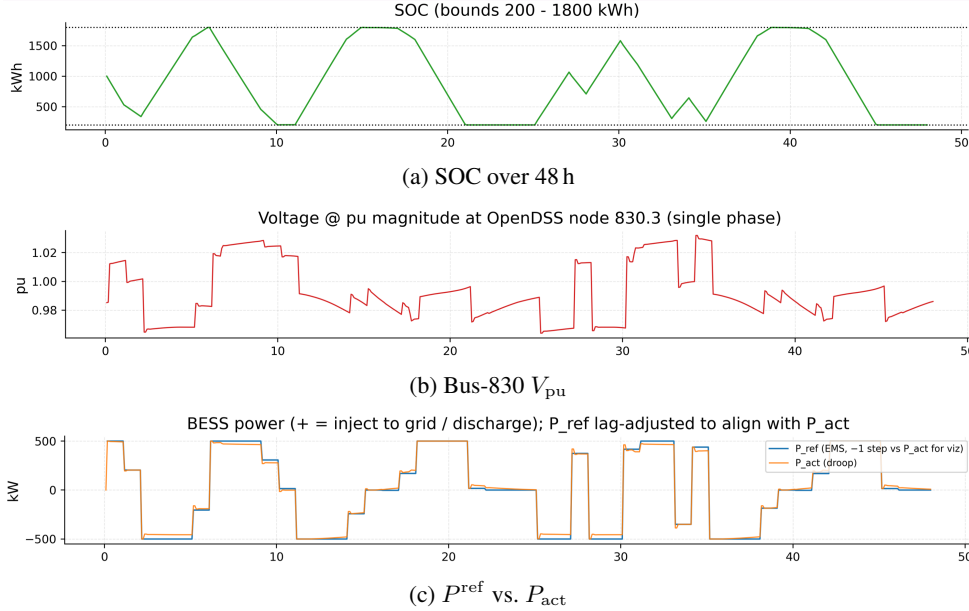


Figure 2: LP-EMS closed-loop validation. 48-h run, $\sigma = 0.004$ \$/kWh (seed 32). SOC cycles within [200, 1800] kWh; voltage stays near 1 pu; droop corrections are small relative to the LP setpoint.

6.2 Behavior Cloning Results

Table 1 summarizes all runs. At $\sigma = 0.004$, BC matched LP revenue exactly (\$6,272) with identical voltage RMSE (0.02304 pu), confirming **H1**: when price uncertainty is minimal and expert-visited states span the operating space, the RF perfectly replicates LP dispatch.

At $\sigma = 0.04$, LP revenue rose to \$13,631 (higher volatility allows more arbitrage); BC captured \$12,291 (90.2%) with setpoint MSE 30,623 kW². At $\sigma = 0.4$, LP reached \$120,712 while BC captured \$108,440 (89.8%) with MSE 39,168 kW². The increasing gap and MSE with σ confirm **H2**.

Notably, BC’s voltage RMSE was *lower* than LP’s at both medium and high noise (0.02320 vs. 0.02496; 0.02389 vs. 0.02523). Because the RF produces smoother P^{ref} than the LP’s aggressive arbitrage, the droop layer has smaller corrections to make and bus voltage fluctuates less.

Table 1: Experimental summary. Revenue R uses $\Delta t_5 = 5/60$ h and unperturbed CSV prices. V-RMSE at bus 830 phase C. MSE compares hourly setpoints against LP under identical conditions. All runs: 40 days, 5-min resolution.

Controller	σ [\$/kWh]	R [\$]	V-RMSE [pu]	MSE [kW ²]
LP-EMS	0.004	6,272	0.02304	—
BC (RF)	0.004	6,272	0.02304	—
LP-EMS	0.04	13,631	0.02496	—
BC (RF)	0.04	12,291	0.02320	30,623
LP-EMS	0.4	120,712	0.02523	—
BC (RF)	0.4	108,440	0.02389	39,168
Dagger	0.4	109,820	0.02391	32,514

6.3 Dagger vs. BC

At $\sigma = 0.4$, DAgger reduced setpoint MSE from 39,168 to 32,514 kW² (17% reduction) and increased revenue from \$108,440 to \$109,820, confirming **H3**. Voltage RMSE was unchanged (0.02391 vs. 0.02389), confirming that the droop deadband and hard SOC clamp provide a safety floor independent of how accurately P_h^{ref} matches the LP.

6.4 Model Architecture Comparison

Among MLP, LSTM, LSTM+auxiliary, and Random Forest, the RF performed best. SOC and the 24-slot price vector form a fixed-width tabular input; RF’s decision-tree ensemble exploits this without modeling temporal dependencies across hours. The LSTM with auxiliary task (predicting all 24 LP-horizon setpoints as auxiliary targets) was the most stable over long runs: even at 40 days it did not collapse under distribution shift—SOC remained dynamic rather than saturating at a boundary.

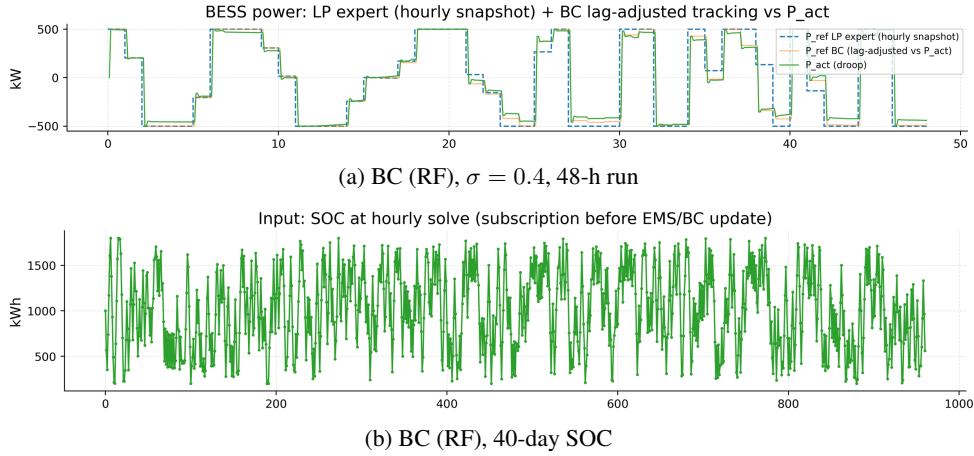


Figure 3: BC (RF) closed-loop results ($\sigma = 0.4$, seed 11). Left: 48-h P^{ref} , P_{act} , and V_{pu} trajectory. Right: 40-day SOC—dynamic throughout, never stuck at a boundary.

6.5 Long-Run Stability

None of the BC models collapsed in 40-day runs. SOC trajectories remained dynamic within [200, 1800] kWh, suggesting the droop federate’s hard clamp ($\text{SOC} \leftarrow \text{clip}(\cdot, 200, 1800)$) prevents runaway divergence. This partially mitigates H2: setpoint MSE grows with σ , but physical bounds limit how severely the policy degrades in realized system behavior.

7 Limitations and Future Work

Several limitations constrain this study. First, we use a single BESS; multi-unit coordination introduces inter-agent coupling. Second, DAgger ran for one iteration; full convergence would likely close more of the LP gap. Third, the LP optimizes only energy arbitrage (SOC dynamics + power bounds); voltage regulation relies entirely on the droop layer, not the LP. Fourth, the price profile repeats the same 24-hour CSV pattern across days; real-world multi-day price variation is not captured. Fifth, the price noise is applied only from simulation hour 24 onwards, meaning the first day is noise-free—a potential in-distribution advantage for BC.

Future directions: (i) multiple DAgger iterations to study convergence; (ii) graph neural network policies that encode feeder topology; (iii) larger feeders (IEEE 123-bus) and multi-BESS settings; (iv) voltage constraints in the LP to compare learned voltage awareness; (v) integration into the outer planning loop to measure wall-clock speedup.

8 Conclusion

We presented a closed-loop imitation learning study for BESS energy arbitrage dispatch in an unbalanced distribution network. The EMS uses a fixed 24-step rolling-horizon LP re-solved each hour; the BC/Dagger policy observes SOC and the rolling price forecast at each hourly boundary and outputs a single active-power setpoint held constant by the droop federate for 12 five-minute sub-steps. A Random Forest trained by BC matched LP revenue exactly at low price noise and retained 89.8% at high noise, achieving *lower* voltage RMSE than LP in all tested conditions. DAgger reduced setpoint MSE by 17% in the hardest scenario. The droop federate’s voltage deadband ($V_{\text{ref}} \pm 0.01$ pu, $K = 2000$ kW/pu) and hard SOC clamping ($[200, 1800]$ kWh) provide a robust safety floor—system-level degradation is bounded even when the learned policy drifts under distribution shift.

Code: <https://github.com/alitasavori/closed-loop-il-battery-control>

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